ARGUS

IMMERSIVE ROBOT CONTROL COCKPIT

Demonstration Safety Report

Info Day 2026 • Public interactive demonstration

EVENT DETAILS

DATE
Saturday, 24 October 2026

LOCATION
Haking Wong Podium

SCOPE
ARGUS cockpit experience, linked robot operation and static robot display

DOCUMENT STATUS
Draft for venue review and pre-event confirmation



Provide a supervised visitor experience while keeping people outside the moving-equipment envelope.

Control principle Supervision • separation • emergency stop • managed visitor flow

1 Event overview and scope

What will operate, what will remain static, and the limits of this safety report.

1.1 Demonstration purpose

ARGUS combines VR, leader arms and foot pedals in a supervised robot-control cockpit. One visitor operates a linked robot at low speed while staff control the segregated robot zone.

LIVE: ARGUS COCKPIT	LIVE: ROBOT 1	STATIC: ROBOT 2
• One seated visitor • Attendant beside participant	• Low-speed linked operation • Robot zone only	• Unpowered display • Stabilised behind barriers

1.2 Safety boundaries

- Covers the cockpit, both robots, power, cables, barriers and visitor area.
- Visitors remain outside the robot zone; live motion is limited to approved tasks.
- Confirm final layout, contacts and venue requirements before opening.

1.3 Personnel and authority

Role	Primary responsibilities
Demonstration supervisor	Approves startup, shutdown and restart; coordinates with venue.
Cockpit attendant	Briefs visitors; monitors comfort and cockpit input.
Robot-zone marshal	Controls robot zone, crowd and emergency access.

Any staff member may stop for zone intrusion, unstable communication, unexpected motion or participant discomfort.

2 System and equipment

Technical information retained from the original report, presented in a consistent operational format.



CONTROL LOOP

- VR, leader-arm and pedal input
- ARGUS command mapping
- Live visual, position and force feedback

2.1 ARGUS immersive cockpit

Approximate mass	20 kg
Approximate dimensions	80 cm × 100 cm × 140 cm (W × D × H)
Power input	One 220 V AC socket supplying a 5 V / 5 A power unit
Operating duration	Continuous while connected to the approved power supply
Primary interfaces	VR headset, leader arms, three-axis foot pedals and operator seat

2.2 Robot 1 — linked bimanual robot

Approximate mass: 15 kg
Approximate height: 120 cm
Power input: Two 220 V AC sockets supplying two 12 V / 2 A power units
Operating duration: Continuous while connected to the approved power supplies
Function: Performs controlled manipulation tasks using robot arms linked to the ARGUS cockpit



2.3 Robot 2 — static showcase platform

Approximate mass: 50 kg
Approximate height: 170 cm
Power and runtime: Not connected; static display only
Function: Introduces the public to the robot platform and its mechanical architecture



3 Operating procedure

A single controlled sequence applies to the cockpit, the linked robot and every visitor session.

3.1 Before opening to visitors

1. Inspect and stabilise the cockpit and robots.
2. Set segregated zones, barriers and clear emergency routes.
3. Secure cables; verify sockets, adapters and enclosures.
4. Test communications, sensors, controls, limits and emergency stops.
5. Run a staff-only low-speed test and brief all staff.

3.2 Visitor-session workflow

1	2	3	4	5
BRIEF	**FIT**	**ENABLE**	**SUPERVISE**	**RESET**
Explain controls, zones and stop signal.	Adjust seat and headset.	Confirm robot zone clear.	Run an approved low-speed task.	Stop, neutralise and sanitise.

3.3 During live operation

- Attendant stays beside the participant with the emergency stop in reach.
- Marshal keeps the robot zone clear and monitors motion.
- Stop for frozen feedback, unstable link, unexpected motion or discomfort.

3.4 Close-down

- Stop motion and isolate power before dismantling.
- Record faults, stops, near misses and incidents.

> Supervisor restart only after the cause is resolved and pre-use checks are repeated.

4 Risk assessment and control measures

Residual risk ratings assume that every listed control is implemented and verified.

Hazard	Persons at risk / consequence	Required controls
Electrical supply and cables	Visitors/staff: shock, burn or trip	Approved adapters; inspect cables; keep liquids away; cover routes; isolate before adjustment.
Cockpit mechanisms	Participant/staff: pinch, impact or strain	One seated user; adjust before enable; brief stop signal; apply motion limits.
VR use	Participant: nausea or disorientation	Seated use; supervise; stop on discomfort; sanitise headset.
Robot 1 workspace	Visitors/staff: impact, trapping or pinch	Barrier beyond maximum reach; low-speed mode; no public entry; tested E-stop.
Control or sensor fault	All persons: unintended motion	Pre-test link and sensors; limit commands; stop on delay, freeze or mismatch.
Equipment instability	Visitors/staff: falling equipment	Level floor; inspect fasteners; stabilise equipment; isolate damage.
Crowd intrusion	Visitors/staff: contact or blocked access	Separate zones; signed barriers; marshal; clear emergency routes.
Manual handling	Staff: strain or crush injury	Power down; plan route; use enough people or aids; lift at approved points.
Weather or unsafe venue	All persons: electrical or slip hazard	Stop live use; isolate equipment; follow venue instructions.

IMPORTANT Reassess risk after any change to layout, task, speed, payload, power or public access.

5 Emergency response, crowd control and weather

Immediate actions are deliberately short so staff can use them under pressure.

5.1 Immediate shutdown sequence

1	Call “STOP” and press the nearest emergency stop.
2	Keep visitors out; keep the participant seated.
3	Isolate power when safe.
4	Provide first aid or call venue emergency services.
5	Secure equipment and prevent reuse.
6	Record the event; supervisor approves restart.

5.2 Scenario-specific response

Scenario	Immediate response	Restart condition
Loss of control or communications	E-stop; isolate robot; hold exclusion zone.	Retest link, sensors and command mapping.
Unexpected motion or failure	Stop, isolate and quarantine. Do not restrain by hand.	Cause identified; inspection passed.
VR discomfort	Stop; remove headset; assist participant.	Participant recovered; briefing reviewed.
Electrical heat, smoke or damage	Stop, isolate if safe and clear the area.	Qualified inspection and venue approval.
Crowd breach	Stop robot; restore barrier and clear zone.	Marshal confirms control.

5.3 Crowd-control layout

- Provide a queue/photo area of at least 2 m × 1 m.
- Separate queue, cockpit and robot zones.
- Mark entry/exit; keep emergency routes clear.
- Post warning signs and assign a marshal.



Reference only: prior supervised public robotics activity.

5.4 Bad-weather arrangement

Stop live operation for rain, water ingress, wind, unsafe flooring or venue instruction. Isolate and secure equipment; use static display only with venue approval.

6 Operational checklists and sign-off

6.1 Pre-opening inspection

☐	Equipment positioned, stable and undamaged.
☐	Barriers, signs and access routes in place.
☐	Power supplies and cables inspected.
☐	Controls, sensors, limits and E-stops tested.
☐	Robot 1 low-speed test passed; Robot 2 unpowered.
☐	Staff roles, contacts and restart rules confirmed.

6.2 Each visitor session

☐	Participant briefed; seat and headset adjusted.
☐	Robot zone clear; marshal ready.
☐	Approved task and low-speed mode selected.
☐	Robot stopped; controls neutral; headset sanitised.

6.3 Close-down

☐	Motion stopped; power isolated.
☐	Faults, stops and incidents recorded.
☐	Equipment secured; damaged items quarantined.